



University of Asia Pacific

Department of Computer Science & Engineering

Course Title : Microprocessors and Microcontrollers Lab

Course Code : CSE 316

Report No : 04

Report Title : Wireless Theft Detection

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1. Experiment Name

Wireless Theft Detection

2. Objective

To design and implement a touch/push-button based door lock system using Arduino, where a servo motor locks/unlocks the door and a buzzer provides different sound indications for each action.

3. Apparatus / Hardware & Software Requirements

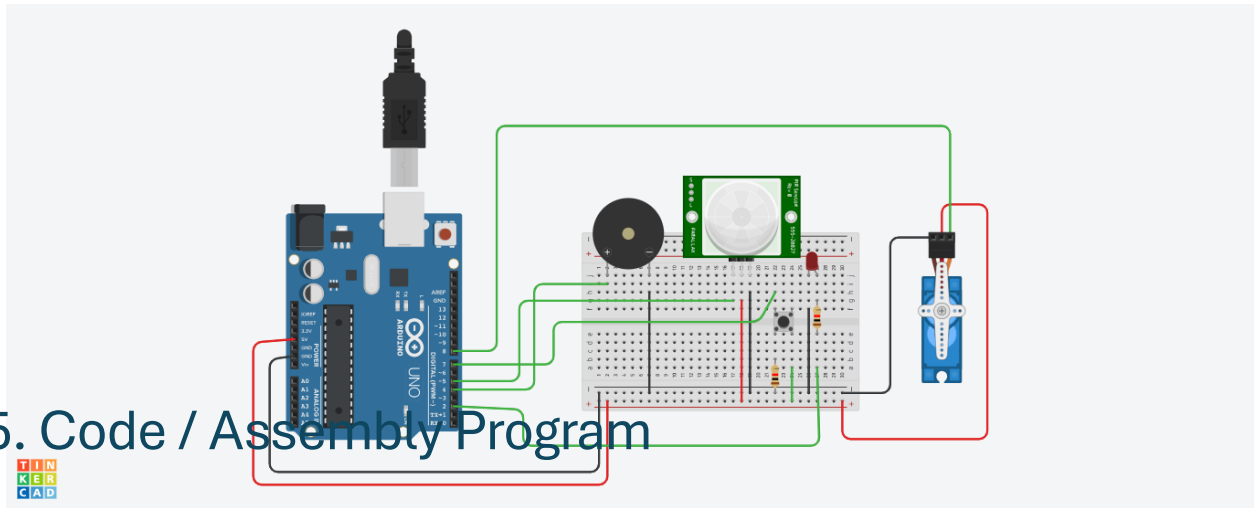
Hardware

- Arduino Uno (or compatible)
- PIR motion sensor (e.g., HC-SR501)
- Servo motor (e.g., SG90)
- Buzzer
- LED + current-limiting resistor (220Ω – 330Ω)
- Push button
- Breadboard and jumper wires
- USB cable / 5V supply

Software

- Arduino IDE
- Arduino Servo library (`#include <Servo.h>`)

4. Circuit Diagram / Schematic



5. Code / Assembly Program

```
#include <Servo.h>

// Pin setup
int pirPin = 5;    // PIR sensor
int buzzerPin = 4; // Buzzer
int ledPin = 2;    // LED
int servoPin = 8;  // Servo
int buttonPin = 7; // Push button
int motionState = LOW;
bool systemOn = false; // Mode OFF initially
Servo myServo;
```

```
void setup() {  
  pinMode(pirPin, INPUT);  
  pinMode(buzzerPin, OUTPUT);  
  pinMode(ledPin, OUTPUT);  
  pinMode(buttonPin, INPUT_PULLUP); // Button  
  
  myServo.attach(servoPin);  
  myServo.write(0); // Initial position  
}  
void loop() {  
  static bool lastButtonState = HIGH;  
  bool currentButtonState = digitalRead(buttonPin);  
  
  if (lastButtonState == HIGH && currentButtonState == LOW)  
  {  
    systemOn = !systemOn; // Toggle mode  
    delay(300);  
  }  
  lastButtonState = currentButtonState;  
}
```

```
if (!systemOn) {  
    motionState = digitalRead(pirPin);  
    if (motionState == HIGH) {  
        // Motion detected  
        digitalWrite(ledPin, HIGH);  
        digitalWrite(buzzerPin, HIGH);  
  
        myServo.write(0); // Rotate servo  
        tone(buzzerPin, 200, 500000);  
        delay(200);  
        noTone(buzzerPin);  
        Serial.println("Motion Detected!");  
    } else {  
        // No motion  
        digitalWrite(ledPin, LOW);  
        digitalWrite(buzzerPin, LOW);  
        noTone(buzzerPin);  
        myServo.write(0); // Back to normal  
    }  
}
```

```

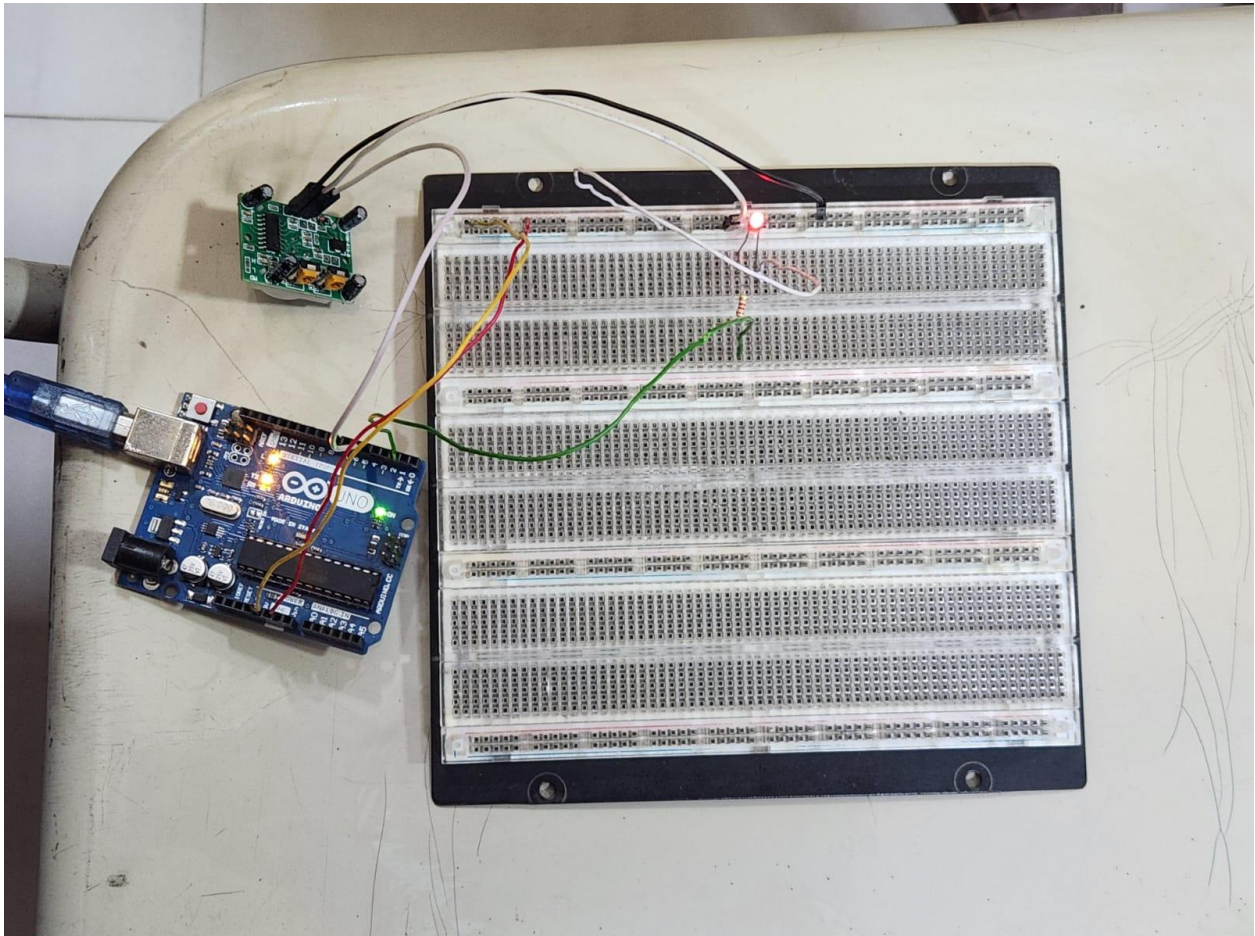
else {
    motionState = digitalRead(pirPin);
    if (motionState == HIGH) {
        // Motion detected
        digitalWrite(ledPin, HIGH);
        myServo.write(90); // Rotate servo
    } else {
        // No motion
        digitalWrite(ledPin, LOW);
        digitalWrite(buzzerPin, LOW);
        myServo.write(0);
    }
}
delay(200);
}

```

6. Output / Observations

- The system has **two modes** controlled by the push button:
 - **Mode 1 (systemOn = false): Alarm mode** (LED + buzzer tone + message)

- **Mode 2 (systemOn = true): Silent mode (LED + servo movement only)**
- Pressing the button toggles the mode; a small delay is used for basic debounce.
- When motion is detected, the PIR sensor output becomes **HIGH**, and the system responds according to the selected mode.



7. Result

After implementing and testing the system, the following observations were recorded:

Alarm Mode (systemOn = false):

When motion was detected, the LED turned ON and the system triggered the buzzer tone (200 Hz), indicating an alert condition. The Serial Monitor (after adding Serial.begin) displayed “Motion Detected!”. When no motion was present, the LED and buzzer turned OFF, and the system returned to standby.

Silent Mode (systemOn = true):

When motion was detected, the LED turned ON and the servo rotated to 90°, indicating a motion-triggered action without sound. When motion was not detected, the LED turned OFF and the servo returned to 0°, keeping the system in a normal idle state.

8. Conclusion

This experiment provided practical understanding of sensor-based automation using Arduino and improved skills in reading a PIR sensor’s digital output to control multiple actuators. It strengthened concepts such as mode selection using a push button (INPUT_PULLUP), conditional logic for

different system behaviors, and coordinating outputs like LEDs, buzzer alerts, and servo positioning.

Overall, the system successfully detected motion and produced correct responses in both modes—an audible alarm mode and a silent servo-based mode. This project forms a strong foundation for future smart security and automation applications such as intrusion alarms, automatic doors, and motion-triggered access control systems.